Reconceptualizing Shadow IT: Motivations, Consequences, and Governance in the Age of Shadow AI

Linus Schärmann

2910412

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Lehrstuhl für Wirtschaftsinformatik
und Systementwicklung
Universität Würzburg

Betreuer: Prof. Dr. Frédéric Thiesse

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Zusammenfassung

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.
Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

Abstract

Obige Zusammenfassung in englischer Sprache.

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List of Abbreviations

AI Artificial Intelligence

BYOD Bring Your Own Device

IT Information Technology

LLM Large Language Model

SAI Shadow AI

SIT Shadow IT

SLR Structured Literature Review

XAI Explainable Artificial Intelligence

1 Introduction - 2 pages

Dieser Teil der Arbeit sollte folgende Inhalte haben: • Einführung in die Problemstellung • Motivation und Herleitung des Themas • Aufbau der Arbeit Hinweis: Es hat sich als hilfreich erwiesen, die Einleitung mit der Zusammenfassung bzw. dem Abstract und der Schlussfolgerung zu vergleichen. Damit stellt man sicher, dass diese inhaltlich im Bezug auf Zielsetzung und Motivation übereinstimmen. Der Umfang sollte ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

2 Theoretical Foundations - 6-8 pages

In order to establish a comprehensive understanding of Shadow IT (SIT) and its further, Shadow AI (SAI), this section provides an overview of the theoretical foundations and definitions relevant to these phenomena.

2.1 Shadow IT

2.1.1 Concept and Definition

SIT is a multifaceted phenomenon encompassing the use, development, or deployment of Information Technology (IT) artifacts (e.g., software, hardware, or services) outside the formal control, approval, or awareness of the central IT function. For instance, Haag and Eckhardt (2017, p. 1) describe it as "hardware, software, or services built, introduced, and/or used for the job without explicit approval or even knowledge of the organization", whereas Kopper and Westner (2016, p. 1) emphasize autonomous deployment within business departments without IT involvement. Other authors similarly highlight the lack of alignment with IT governance (Käss et al., 2021, p. 4; Rakovic et al., 2020, p. 4).

Beyond these operational perspectives, SIT can be conceptualized as a structural complement to formal IT, consisting of independently developed and used systems, processes, and practices that are not known, supported, or accepted by the official IT department (Rakovic et al., 2020, p. 147). More concise definitions further highlight either the user-driven development outside central IT or the absence of approval and monitoring as defining features (Rakovic et al., 2020, p. 4; Magunduni and Chigona, n.d., p. 2).

Collectively, these definitions establish SIT as a theoretically grounded construct characterized by autonomy, decentralization, and non-sanctioned IT activity, emerging outside formal IT governance.

2.1.2 Scope

SIT encompasses a wide range of tools and solutions, including spreadsheets, databases, cloud services, mobile devices, and other ad-hoc IT solutions (Rakovic et al., 2020, p. 2; Kopper and Westner, 2016, p. 1). These systems are typically small-scale, developed by individuals or small teams, and tailored to immediate operational needs (Rakovic et al., 2020, p. 2). Their emergence reflects both the accessibility of modern IT tools and the flexibility required by employees in dynamic work environments.

2.1.3 Conceptual Distinctions

While SIT shares similarities with related practices such as workarounds, and Bring Your Own Device (BYOD) programs, it is distinct in being unauthorized and unsanctioned. Workarounds represent behavioral adaptations to organizational constraints, whereas SIT manifests these adaptations through technological artifacts. IT consumerization, on the other hand, refers to the broader trend of user-driven technology adoption, providing a context in which SIT can emerge (Rakovic et al., 2020, p. 2; Haag and Eckhardt, 2017, p. 2).

2.1.4 Core Characteristics

SIT is not necessarily malicious or risky in intention. Users often deploy it to enhance productivity, overcome system limitations, or facilitate collaboration. Its defining features are low visibility, lack of IT support, and a tension between organizational control and individual initiative.

2.2 Shadow AI

Definition of Shadow AI

How Shadow AI differs from Shadow IT

Unique Characteristics

Examples

Emergence and Growth

Scale and Prevalence

Sociotechnical Nature and Governance Challenges

Knowledge Leakage and Data Risk

Governance Drift and Policy Gaps

Organizational Context

Attack Surface and Technical Risks

Ethical Paradox

2.3 Conceptual Background on Shadow Behavior

This section should: • briefly explain why shadow behavior exists in general • stay high-level and theory-driven • avoid listing concrete motivations

Example content: • Workarounds in organizations • Tension between control and autonomy • Informal vs. formal IT usage • Knowledge worker behavior

3 Methodology - 2-3 pages

The review follows a structured approach based on the guidelines provided by Okoli (2015, pp. 887–902) for conducting a Structured Literature Review (SLR) in the field of information systems research. The methodology is organized into four phases of two steps each: Planning, Selection, Extraction, and Execution (Okoli, 2015, p. 885).

While the original framework is designed for collaborative research settings involving multiple reviewers, this review was conducted by a single researcher. Therefore, minor adaptions were made to certain steps (e.g., drafting the research protocol, team training) to ensure methodological rigor while maintaining the integrity of the review process. The following sections describe these phases and their implementation in the context of this review.

3.1 Purpose

The first step in conducting a SLR is to define the purpose of the review, which is to identify and analyze existing literature on the phenomenon of SIT and further the development of SAI in organizational contexts. The SLR aims to synthesize insights on the motivations behind SIT and SAI, their organizational consequences, and the governance approaches employed to manage these phenomena.

3.2 Draft Protocol

The second step includes drafting a research question, a research protocol, and training the team. Since this thesis is conducted by a single person, the protocol and team training steps are not required and will therefore not be carried out further. The research questions guiding this review are as follows:

Main Research Question (RQ): How does SAI differ from traditional SIT in organizations with regard to motivational factors, consequences, and countermeasures?

To better address the research question, it will be broken down into three sub-questions and answered throughout the paper:

1. Which motivational factors known from SIT are transferable to SAI, and which additional motivation factors are specific to SAI?
2. Which organizational consequences of SAI differ from those of SIT?

3. Which countermeasures are discussed for SIT and SAI, and how do they differ in design and expected effectiveness?

3.3 Practical Screening

To make sure only relevant and high-quality literature is included in the review, practical screening criteria are defined to filter the large number of initial search results before reviewing them in detail.

For this review, only journal articles, conference proceedings, and book chapters/sections published in English between 2005 and 2026 with a focus on SIT/SAI in organizational or professional settings were included. The research design of the included studies could be empirical (quantitative, qualitative, or mixed methods), theoretical, conceptual, or review-based, as long as they provide insights relevant to the research questions. The participants in the included studies could be professionals, managers, employees in various roles (e.g., analysts, data scientists, IT staff), or end-users as long as they have a close connection to the organizational context of SIT and SAI.

3.4 Literature Search

The literature search was conducted using four search strings across four databases (AISeL, EBSCO, IEEE, and Web of Science), employing advanced search with Boolean operators (`OR` and `AND`) applied to all fields or title/abstract/keywords depending on the database. The search strings were designed to capture literature on both concepts, as well as their intersection, to ensure a broad and comprehensive coverage of relevant studies. The search strings and their purposes are summarized in Table 1. The search process was completed on March 15, 2026, and therefore includes only literature published up to that date.

Table 1: Literature Search Strings

String	**Purpose**	**Query**
S1	Shadow IT Baseline	`"shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system"`
S2	Shadow AI Baseline	`"shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI"`
S3.1	Shadow IT × Shadow AI	`("shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system") AND ("shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI")`
S3.2	Shadow IT × AI (broader)	`"shadow IT" AND "artificial intelligence"`

String S3.2 was included to capture literature not explicitly using the term *Shadow AI* but discussing the intersection of SIT and Artificial Intelligence (AI) technologies.

Additionally, the search was extended by backward and forward searching of the reference list of the included studies to identify any relevant literature that may have been missed in the initial database search.

3.5 Data Extraction

The remaining literature was subjected to a systematic data extraction process, where the literature was coded and sorted based on their key scope in accordance to the research questions. Based on this preliminary coding, relevant information was extracted and collected in a structured format to group similar concepts and themes together. This process was conducted iteratively, with multiple rounds of coding and extraction to ensure that all relevant information was captured and organized effectively for further analysis and synthesis in the subsequent phases of the review.

3.6 Quality Appraisal

While the practical screening criteria help filter out irrelevant studies before even reading the full-text, the review process also involves a quality appraisal to ensure the included studies meet certain standards. Therefore exclusion criteria need to be defined and applied to the extracted studies (Okoli, 2015, p. 884).

This SLR excluded studies with unsufficient theoretical contribution, unclear methodology or findings, or low prerequisites to adequately answer the research questions.

3.7 Study Synthesization

Study synthesization involves aggregating, summarizing, and interpreting the findings from the included studies to answer the research questions (Okoli, 2015, p. 899). The aggregation utilized the preliminary outline of the review, which was based on the research questions, to group similar concepts and themes, such as definitions, motivations, consequences, and countermeasures of SIT and SAI. [1]

3.8 Writing the Review

The final phase of the review process presents the results in detail, so future reviewers can understand and reproduce the process and the findings (Okoli, 2015, p. 901). Section 4 presents the results of the SLR, organized around the key themes of identified lit-

[1] To support the aggregation and summarization of relevant data, generative AI tools such as ChatGPT (OpenAI, GPT-5.3, April 2026 version) and Claude (Antrophic, Sonnet 4.5, April 2026 version) were used. All outputs were critically reviewed and revised by the author who assumes full responsibility for the final content.

erature, motivational factors, organizational consequences, and governance approaches of SIT and SAI to summarize existing literature and knowledge about these concepts.

4 Structured Literature Review Results - 8-10 pages

4.1 Overview of Identified Literature

In total, 831 records from the four databases were identified. Table 2 provides a breakdown of the search results per string and database, illustrating the initial scope of literature available for review.

Table 2: Database Search Results per String

String	AISeL	EBSCO	IEEE	WoS	Total
S1	296	227	39	94	656
S2	6	79	9	5	99
S3.1	4	2	0	0	6
S3.2	45	18	4	3	70
Total	**351**	**326**	**52**	**102**	**831**

Following the removal of duplicates (124 records) and the application of the practical screening criteria (69 records), 638 records remained for content screening. Reviewing titles and abstracts led to the exclusion of 516 records, leaving 122 records for full-text review and quality appraisal. The full-text review resulted in the exclusion of 41 studies, with 19 additionally excluded due to inaccessibility. The forward and backward searching of references added seven more studies, leading to a final set of 69 included studies for this review.

Table 3 provides a detailed overview of the whole SLR process, including the number of records included, excluded, and remaining at each stage of the screening funnel.

Figure 1 illustrates the temporal distribution of the included studies by publication year, differentiating between SIT and SAI. The results show that SIT has been an established research stream since atleast 2012, with continuous presence and fluctuating publication activity over time, peaking in 2019. In contrast, publication on SAI only begin to emerge in 2021 and remain limited in number. A noticeable increase can be observed from 2024 onwards, with a peak in 2025. Despite this recent growth, the overall volume of SAI remains substantially lower compared to SIT. Overall, the figure highlights that while SIT represents a mature and consistently researched phenomenon, SAI is a comparatively developing area of research.

Table 3: Literature Screening Process

Stage	**Included**	**Excluded**	**Remaining**
Records identified	–	–	831
After duplicate removal	–	124	707
Practical screening			
Language (English only)	–	24	683
Publication year (2005–present)	–	30	653
Publication type	–	15	638
Content screening			
Title screening	254	384	254
Abstract screening	122	132	122
Full-text review			
Not accessible	–	19	103
Full-text screening	62	41	62
Forward searching	3	–	65
Backward searching	4	–	69

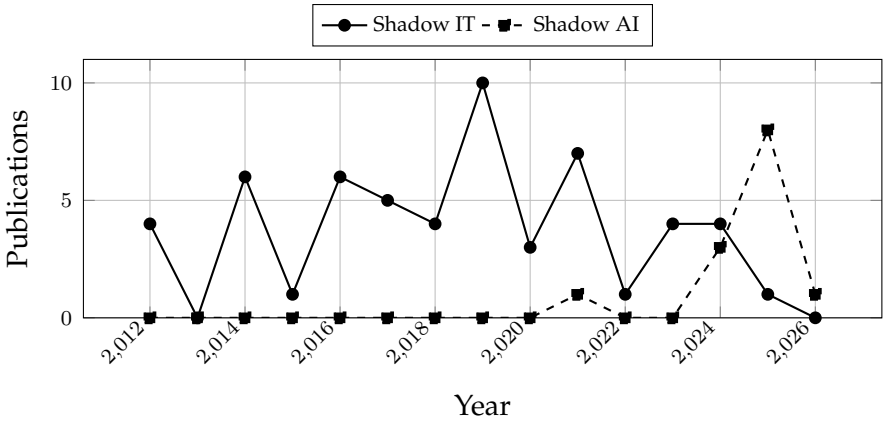

Figure 1: Temporal distribution of included studies on Shadow IT and Shadow AI (2012–2026)

4.2 Motivational Factors

The following presents the different orientations of motivational factors.

4.2.1 Shadow IT

4.2.2 Shadow AI

The literature on SAI suggests that the unauthorized usage of AI tools is rarely driven by a desire to cause harm. Instead, it is characterized as a functional, user-driven response to specific workplace demands where official IT protocols are avoided for personal and professional profit (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 3). This section explores the multi-faceted motivations behind this behavior, ranging from individual productivity needs to systematical organizational failures.

The most dominant driver identified across the literature is the immediate pursiut of efficiency. Employees frequently describe AI-powered tools as "game changers" for automating repetitive tasks and streamlining workflows (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 9). This behavior is often percieved as beneficial by the users because it improves overall efficieny and provides analytical insigts that sanctioned tools may lack (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 7).

In practical application, generative AI is adopted to achieve significant time-saving enhancements in several core areas:

- **Communication and Synthesis:** Employees use tools to draft emails, summarize lengthy documents, and brainstorm new ideas (Ross, Hibbert, and Moss, 2025, p. 4).
- **Decision Augmentation:** AI is leveraged to assist in decision-making, information processing, and coordination complex tasks (Alter, 2024, p. 11).
- **Routine Automation:** Workers turn to these tools because they significantly enhance their productivity by automating routine tasks and accelerate human decision-making (Ross, Hibbert, and Moss, 2025, p. 3).

Research indicates that the "pressing demand for productivity gains" is a primary motivator for employees (Ross, Hibbert, and Moss, 2025, p. 2). This is especially true under high task presure, where usage becomes almost entirely efficiency-driven (Raizada, Shakya, and Rahul, 2026, p. 1140). Furthermore, the psychological safety and percieved productivity benefits often outweigh compliance concerns for the average worker (Raizada, Shakya, and Rahul, 2026, p. 1136). Thus, employees seek out third-party AI tools even in sensitive industries like healthcare to fill gaps left by official resources (Balogun et al., 2025, p. 2; Varshini and Sampathkumari, 2025, p. 2).

Another significant factor for the adoption of SAI is the frustration stemming from traditional IT management. Many employees turn to unsactioned tools as a direct response to percieved inefficiencies in their organization's sanctioned digital infrastructure (Ross, Hibbert, and Moss, 2025, p. 2).

Specific organizational drivers include:

- **Bureaucratic Friction:** Adoption is often fueled by bureaucratic bottlenecks, unresponsiveness from centralized systems, and a misalignment between official IT infrastructure and actual operational needs (Ross, Hibbert, and Moss, 2025, p. 1).
- **Resource Scarcity:** The informal adoption of AI is frequently an operational necessity born from limited technical resources (Ross, Hibbert, and Moss, 2025, p. 1).
- **Governance Failure:** A "governance drift zone" often exists where formal policies are present but lack real-world traction, leading employees to find their own solutions (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 1).

When faced with bureaucratic inertia or a lack of accessible, sanctioned alternatives, workers feel compelled to bypass traditional IT procurement to adopt sophisticated Large Language Model (LLM)s for their daily tasks (Ross, Hibbert, and Moss, 2025, p. 6; Raizada, Shakya, and Rahul, 2026, p. 1136). This creates a situation where employees who use unauthorized AI report different perceptions of task burden and work pressure compared to those who strictly follow protocols (Varshini and Sampathkumari, 2025, p. 5).

Beyond simple efficiency, SAI is driven by a desire for professional agility and the freedom to innovate. Users claim that leveraging these technologies enables them to be more innovative at work (Chin et al., 2025, p. 2). For many, AI has "sparked creativity", allowing them to approach problem-solving from new perspectives (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 9).

The "innovation-seeking" behavior creates a workplace culture where the agility gained through experimentaion with AI is seen as a justification for bypassing slow, official approval processes (Raizada, Shakya, and Rahul, 2026, pp. 1136, 1138). In industries like healthcare, finance, and education, the drive for innovation and agility is a core reason for seeking alternative tools (Balogun et al., 2025, p. 16). This culture of "digital improvisation" allows innovation to effectively "bypass control" often driving usage further underground once the tools are banned (Ross, Hibbert, and Moss, 2025, p. 1; Raizada, Shakya, and Rahul, 2026, p. 1136).

SAI also serves as a strategic tool for individual and team survival. At a technical level, employees use AI skills to compensate for specific technical weaknesses (An, Lin, and Luo, 2024, p. 3). Teams often resort to consumer-graade AI solutions to actively bridge existing skill gaps (Ross, Hibbert, and Moss, 2025, pp. 1–2).

From a competitive standpoint, the adoption of these tools is viewed as a way to create new opportunities for competitive advantage (Puthal, Ray, and Mohanty, n.d., p. 973). In the modern digital economy, characterized by rapid technological innovation and globalizationm mnany employees feel it is inevitable and necessary for the firm's long-term survival, to adopt AI technology (An, Lin, and Luo, 2024, p. 1). Key user motivations are often linked to a desire for responsible innovation and skills development that the organization may not yet formally support (Anthuvan et al., 2025, pp. 2708–2709).

Finally, the absence of clear policy often acts as a silent enabler. Many employees hide their AI usage specifically because formal guidance or policy is lacking (Varshini and Sampathkumari, 2025, p. 2). This combination of low policy awareness and vague approval mechanisms facilitates risk-taking behavior among employees (Raizada, Shakya, and Rahul, 2026, pp. 1138, 1140). In many cases, shadow systems emerge as a direct result of the policy absence, IT bottlenecks, and general employee frustration (Raizada, Shakya, and Rahul, 2026, p. 1138).

4.3 Organizational Consequences

4.3.1 Shadow IT

4.3.2 Shadow AI

While the adoption of SAI is often driven by the pursuit of individual efficiency, its unsanctioned nature introduces a wide array of organizational, technical, and ethical repercussions. The literature indicates that the consequences of these practices extend far beyond simple policy violations, impacting the core security architecture, legal standing, and cultural fabric of contemporary organizations.

The most immediate technical consequence of SAI is the introduction of systemic vulnerabilities that fundamentally challenge traditional cybersecurity models (Ross, Hibbert, and Moss, 2025, p. 1). By bypassing formal vetting processes, these tools significantly expand the attack surface available to cybercriminals, making their presence a critical concern for security professionals (Balogun et al., 2025, p. 5).

Specific technical risks identified in the literature include:

- **Adversarial Exploitation:** There is a growing risk of "shadow backdoor models" being engineered specifically to facilitate adversarial attacks against the organization (Puthal, Ray, and Mohanty, n.d., p. 1).
- **Operational Disruptions:** The proliferation of unsanctioned systems poses risks that range from data breaches to significant operational disruptions (Whitford, Marcel, and McCall, 2025, p. 1).
- **Infrastructure Fragility:** Because these tools are not integrated into official monitoring systems, they create "blind spots" that undermine the overall resilience of the corporate IT environment (Ross, Hibbert, and Moss, 2025, p. 1).

SAI usage frequently results in the erosion of data privacy and the violation of complex regulatory frameworks. A primary concern is "data leakage", where sensitive business information is inadvertently shared with unauthorized recipients through third-party AI models (Puthal, Ray, and Mohanty, n.d., p. 1; Raizada, Shakya, and Rahul, 2026, p. 1).

The consequences for data sovereignty are particularly severe:

- **Legal Violations:** Unauthorized AI usage often breaks data residency and sovereignty

laws, as sensitive information is frequently moved to and stored in jurisdictions that do not meet required security standards (Puthal, Ray, and Mohanty, n.d., p. 2).

- **Regulatory Non-Compliance:** Unmonitored use of Large Language Models (LLMs) leads to documented "compliance breaches" that can trigger significant legal penalties (Raizada, Shakya, and Rahul, 2026, p. 1).
- **Knowledge Leakage:** Beyond individual data points, there is a broader risk of Organizational Knowledge Leakage (OKL), where proprietary intellectual property and strategic insights are exposed through employee interactions with external AI (Chin et al., 2025, p. 1).

The social and cultural repercussions of SAI are equally significant, often manifesting as a breakdown in institutional trust. The literature highlights that unauthorized use has profound repercussions on team dynamics and the overall trustworthiness of the workplace environment (Varshini and Sampathkumari, 2025, p. 1).

The phenomenon of "Bring Your Own AI" (BYOAI) is noted for creating persistent tensions between the desire for innovation, the necessity of trust, and the requirements of regulatory oversight (Anthuvan et al., 2025, p. 1). This often leads to a "governance drift zone", a state where formal policies exist on paper but lack any real-world traction or influence over employee behavior (M. Silic, D. Silic, and Kind-Trüller, 2025, p. 1). This drift creates a disconnect between management and staff, potentially fostering a culture of secrecy rather than collaborative innovation.

A critical consequence in high-risk fields is the emergence of responsibility and accountability gaps. Because many SAI systems function as "black boxes", it becomes nearly impossible for organizations to determine exactly how certain decisions were reached (Balogun et al., 2025, p. 4; M. Silic, D. Silic, and Kind-Trüller, 2025, p. 1).

This lack of transparency exposes institutions to significant legal liability, particularly when AI-driven decisions lead to negative outcomes. In these scenarios, the inability to provide a clear audit trail or explanation for an AI's output makes it difficult to assign responsibility, leaving the organization legally vulnerable (Balogun et al., 2025, p. 4).

The consequences of SAI are particularly acute in sensitive sectors where the margin for error is minimal:

- **Healthcare:** The use of unregulated AI diagnostic tools can lead to misdiagnoses, malpractice, and catastrophic breaches of patient confidentiality (Balogun et al., 2025, p. 2).
- **Finance:** In the financial sector, unregulated AI-driven credit scoring systems have already resulted in discriminatory lending practices, subsequently prompting intense regulatory scrutiny (Balogun et al., 2025, p. 3).
- **Education:** Within academic environments, unauthorized AI applications contribute to academic dishonesty, flawed grading mechanisms, and the violation of student privacy rights (Balogun et al., 2025, p. 2).

Finally, there is a broader societal consequence: the unchecked expansion of SAI risks entrenching systemic biases. If left unmanaged, these tools may further marginalize vulnerable populations by automating and scaling discriminatory logic that has not been subjected to ethical review (Balogun et al., 2025, p. 3). In addition to bias, organizations must contend with "model hallucination", where AI produces confident but entirely false information, leading to flawed outputs and compromised decision-making (Raizada, Shakya, and Rahul, 2026, p. 1).

4.4 Governance Approaches and Countermeasures

4.4.1 Shadow IT

4.4.2 Shadow AI

As organizations recognize that total prohibition of SAI is often impractical and counterproductive, the literature has shifted toward a more nuanced spectrum of countermeasures. These range from high-level strategic frameworks to granular technical monitoring, all aimed at transitioning from a state of "shadow" usage to a sanctioned, transparent environment.

The first line of defense against the risks of SAI involves technical visibility. Organizations are increasingly evaluating the effectiveness of automated monitoring and anomaly detection to identify unsanctioned AI traffic (Whitford, Marcel, and McCall, 2025, p. 1). Effective risk management requires the integration of "AI-aware" safeguards, specifically Data Loss Prevention (DLP) tools, content filtering, and dedicated AI traffic monitoring (Puthal, Ray, and Mohanty, n.d., p. 2).

A fundamental technical countermeasure is the implementation of automated AI discovery and inventory systems (Puthal, Ray, and Mohanty, n.d., p. 1). These systems allow IT departments to maintain a clear record of which tools are being accessed, which is a prerequisite for applying any advanced governance policy. By providing a real-time overview of the organization's AI footprint, these tools help eliminate the "blind spots" traditionally associated with shadow systems.

Beyond simple monitoring, the literature proposes structural changes to how AI assets are handled within the corporate network. One of the most advanced technical approaches is the adoption of a Zero Trust Architecture (Puthal, Ray, and Mohanty, n.d., p. 1). Under this model, every AI asset—regardless of whether it is internal or a third-party application—is treated as a potential threat. This requires constant validation and rigorous access control for every single interaction (Puthal, Ray, and Mohanty, n.d., p. 1).

Furthermore, the "black box" nature of many SAI tools is addressed through the promotion of Explainable Artificial Intelligence (XAI). XAI serves as a critical countermeasure by making AI decision-making processes interpretable for human supervisors (Balogun et al., 2025, p. 7). This transparency allows organizations to identify and rectify biases

or errors in AI outputs before they result in operational or legal harm (Balogun et al., 2025, p. 7).

The literature emphasizes that technical tools must be embedded within broader strategic governance frameworks. For instance, the **BYOAI-GOVT** framework offers a behavior-aware model that categorizes tasks into "risk zones": *Enable*, *Regulate*, and *Restrict* (Anthuvan et al., 2025, p. 1). This framework allows for a tiered response to AI usage based on a four-level maturity scale, moving away from binary "yes/no" policies (Anthuvan et al., 2025, p. 1).

Other researchers suggest that a successful governance structure must include:

- **Risk Classification:** Developing a typology of SAI use cases to understand and rank specific threats (Ross, Hibbert, and Moss, 2025, p. 1).
- **Organizational Bodies:** Establishing AI ethics boards and dedicated compliance units to oversee implementation (Balogun et al., 2025, p. 3).
- **Continuous Auditing:** Implementing regular AI compliance audits and risk assessments to identify emerging vulnerabilities (Balogun et al., 2025, p. 7).

Because SAI is fundamentally a human behavior, several countermeasures focus on education and culture. Employee training programs are deemed essential for raising awareness about the dangers of unauthorized AI and reinforcing cybersecurity policies (Balogun et al., 2025, p. 6). In academic and research settings, this extends to "AI literacy programs" designed to ensure that the use of generative tools does not compromise academic integrity (Balogun et al., 2025, p. 6).

However, education alone is insufficient if it is not supported by the organizational culture. A digital organizational culture plays a vital "positive moderating role," helping to align individual AI skills with authorized organizational innovation (An, Lin, and Luo, 2024, p. 1). Management must address cultural challenges, such as employee resistance to change and deep-seated ethical concerns, to ensure that new governance policies are actually followed rather than circumvented (Murire, 2024, p. 1).

Finally, internal governance is increasingly being dictated by external regulatory requirements. The literature points to the **European Union's Artificial Intelligence Act** as a primary driver for compliance, as it categorizes AI applications by risk level and imposes strict legal obligations on organizations (Balogun et al., 2025, p. 7). Adhering to established standards, such as those provided by the National Institute of Standards and Technology (NIST), is recommended as a best practice for conducting thorough risk assessments (Balogun et al., 2025, p. 7).

Furthermore, there is a call for regulatory agencies to take a more active role by strengthening oversight and imposing stricter penalties for the unauthorized use of AI, particularly in high-stakes industries (Balogun et al., 2025, p. 3). By combining legal pressure with internal auditing, organizations can create a more robust environment where the benefits of AI are realized without the risks of shadow implementation.

5 Conceptual Development: Extending Shadow IT to Shadow AI - 4-5 pages

REPHRASE

Building on the previously established theoretical foundations and the insights of the SLR results, this section focuses on the conceptual linkage and analytical extension of SIT to SAI. While SIT has been extensively studied over the past decades, resulting in a well-established understanding of its underlying motivations, organizational consequences, and governance approaches, the emergence of SAI presents a relatively novel phenomenon. Existing literature on SAI is still limited in scope and fragmentation, offering only initial insights withoout a comprehensive conceptual framework.

To address this gap, this section adopts a transfer-oriented perspective. It systematically draws on the established body of knowledge on SIT and examines how its core concepts, theories, and findings can be applied to the context of SAI. This approach is based on the assumption that SAI, while technologically more advanced, represents an evolution rather than a completely independent phenomenon. As such, similar organizational drivers and bevavioral patterns are expected to persist, even though they may manifest differently due to the specific characteristics of AI technologies.

At the same time, the transfer is not assumed to be one-to-one. AI systems introduce new properties, such as increased autonomy, opacity, and scalability, that may alter both the nature and the impact of associated risks and opportunities. Therefore, the analysis not only seeks to identify parallels between SIT and SAI but also to highlight important deviations that emerge in the AI context.

By structuring the conceptual development along the dimensions of motivations, consequences, and governance approaches, this section provides a theoretically grounded foundation for the subsequent analysis. It enables a systematic exploration of SAI despite the current limitations of dedicated literatuer, while ensuring that the insights remain anchored in established research.

REPHRASE

5.1 Transferable Motivations

5.2 Novel Motivations

5.3 Behavioral Shift in Knowledge Work

5.4 Integrated Conceptual Framework

6 Comparative Analysis of Shadow IT and Shadow AI - 3-4 pages

6.1 Differences in Motivational Drivers

6.2 Differences in Organizational Risk

6.3 Differences in Governance Approaches

7 Discussion - 3-4 pages

8 Limitations

While this review provides a comprehensive overview and valuable conceptual insights into motivations, consequences, and governance approaches of SAI as a emerging topic, it is not without limitations.

First, the literature search was limited to four databases and specific search strings, which may have resulted in the exclusion of other relevant studies that use different terminology or appear in journals not covered by the databases. Although backward and forward searching was conducted to mitigate this risk, it cannot be completely ruled out.

Second, the review was conducted by a single researcher, which may introduce potential bias in the selection, coding, and interpretation of the literature. While efforts were made to ensure rigor and objectivity, the absence of multiple reviewers may limit the reliability of the presented fidings.

Third, the conceptual development and comparative analysis are based on the assumption of a transferability of insights from SIT with the risk of oversimplification in the context of AI due to the different technological characteristics and types of AI tools.

Fourth and finally, the review is limited to literature published to the current date (March 2026), which may not capture all relevant facets of the emerging field of SAI.

9 Future Research Directions

Given the findings and limitations outlined above, there is ample scope for future research on this topic. Empirical studies are needed to validate the transferability of SIT to SAI and to explore the AI-specific characteristics and their impact on organizational processes and outcomes. Qualitative studies, such as case studies or interviews, could provide deeper insights into the changes in organizational practices and employee behavior, while quantitative studies could test the relevance and strength of identified relationships.

Although many forms of SAI can be derived from and build upon the foundations of SIT, AI poses some unique risks that have either already been studied, have not yet been identified, or have yet to emerge. For this reason, more in-depth and specific long-term research is needed.

Overall the field of SAI is still in its early stages, calling for a wide range of research areas (e.g. socio-technical, cybersecurity, ethics) to be explored.

10 Conclusion - 2 pages

In der Schlussfolgerung sollen

• die Themenstellung • der gewählte Ansatz • die Ergebnisse der Arbeit • eine kritische Stellungnahme/Einschätzung • nächste Schritte

deutlich werden. Hinweis: Die Schlussfolgerung sollte mit der Zusammenfassung bzw. dem Abstract und der Einleitung abgeglichen werden. Es sollte immer eine Zusammenfassung der wesentlichen Erkenntnisse der eigenen Arbeit sein, die den Forschungsbeitrag darstellt. Der Umfang der Schlussfolgerung sollte ähnlich wie die Einleitung ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

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A Appendix A